

STAT 408 Lab 6: Maps

Feb 18, 2026

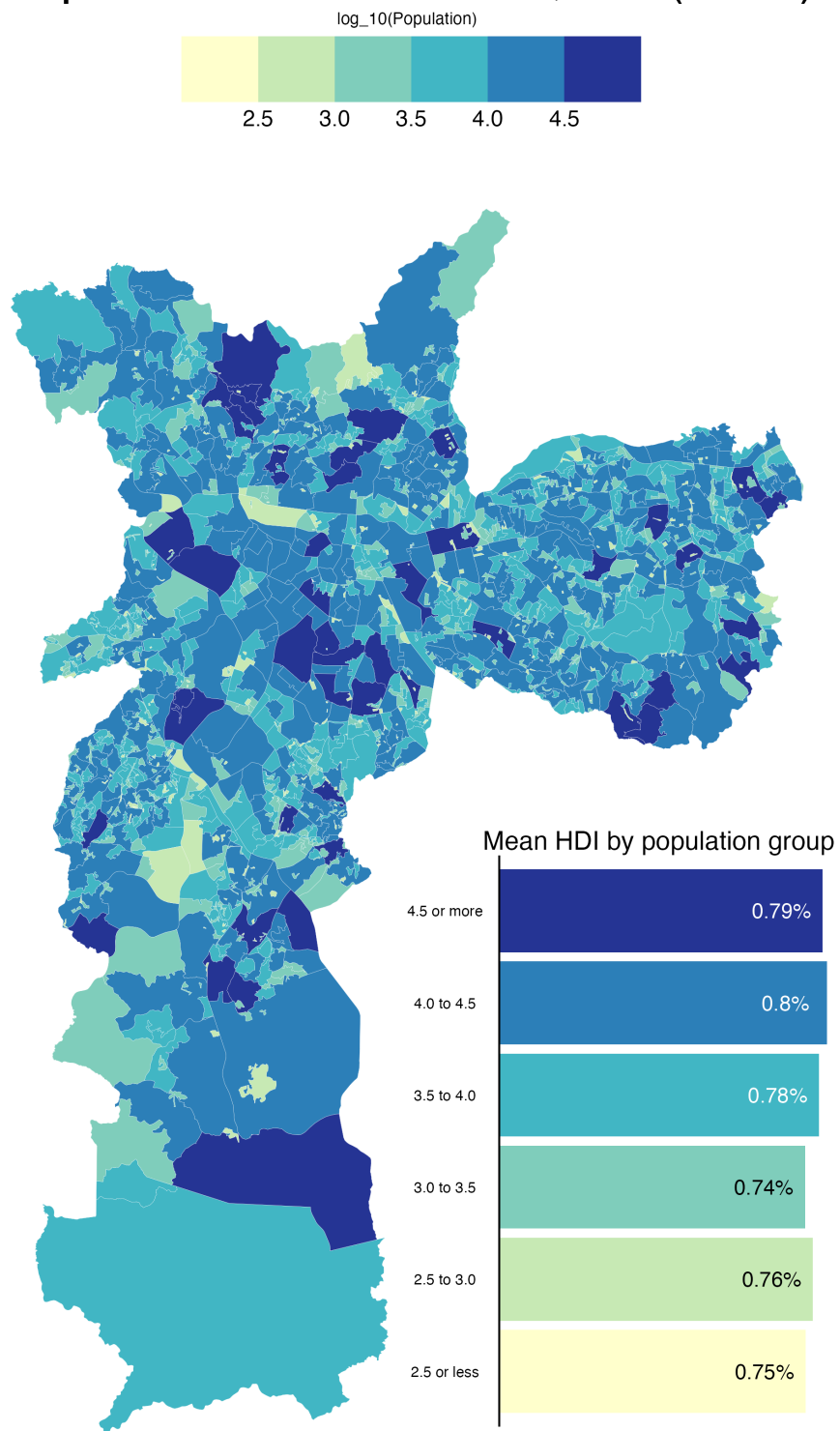
Create a new quarto document titled “STAT 408 Lab 6: Maps” and put your name and some sensible date (the due date or today’s date, for instance). Include **just the code needed to make the plot on the next page** (no extraneous code), **show the plot it generates**, and **use a text paragraph (not just code comments) to summarize the changes you made and why you made them**.

To make this plot, you’ll be able to use the data and mostly reuse the code from `lab06_chloropleth_with_barchart.R`, with some modifications:

- The map should show $\log_{10}(\text{Population})$ instead of HDI
- The barchart should show average HDI for each of the color bins you use for the log population
- The titles, labels, text, formatting, etc. should be adapted accordingly to make the plot readable and pleasing to the eye.

Some bonus points are possible if you improve the plot beyond what is shown below, such as replacing `log_10` in the subtitle with $\log_{10}$.

Population in Sao Paulo, BR (2010)



Source: Atlas Brasil